Chaos Language Algorithm: Technical Summary of Five Standard Attractor System Benchmarks

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Abstract

The Chaos Language Algorithm (CLA) converts chaotic dynamical trajectories into symbolic strings and then learns compact grammar productions via n-gram chunk mining, context-category induction, and minimum-description-length (MDL) greedy acceptance. We report benchmark results on five standard attractor systems—Lorenz-63, Rössler, Mackey-Glass, Lorenz-96, and the logistic map—with exact reconstruction verified in every case. Dimensionality scaling is explored via Lorenz-96 at dimensions 5, 8, 12, 16, 20, and 24, covering the range from textbook systems to OmegaSim-scale state vectors (∼16 continuous dimensions).

1 Algorithm Overview

The CLA pipeline consists of four stages:

1. **Trajectory generation.** Deterministic, fixed-step integrators (RK4 for ODE systems, Euler with discrete delay buffer for Mackey-Glass, direct iteration for the logistic map) produce a sequence of state vectors.
2. **M1 symbolization.** Each trajectory point is mapped to a symbol via fixed rectangular partitioning. For an n-dimensional trajectory, the symbol is a composite of per-axis bin indices: `m1:d0|d1|...|`$\mathtt{d}_{n-1}$. Bounds are inferred once from the full trajectory and held fixed.
3. **Grammar learning.** An n-gram pattern miner discovers repeated substrings of length 2–5. A greedy MDL loop iteratively accepts the chunk or category edit that maximally reduces total description length. Context categories group symbols by co-occurrence using Jensen-Shannon divergence.
4. **Verification.** The learned grammar is expanded back to the original symbol stream. If the expansion matches exactly, the run passes.

2 Benchmark Systems

2.1 Lorenz-63

The classic 3D convection model:

$$\dot{x} = \sigma(y - x) \tag{1}$$
$$\dot{y} = x(\rho - z) - y \tag{2}$$
$$\dot{z} = xy - \beta z \tag{3}$$

with canonical parameters $\sigma = 10$, $\rho = 28$, $\beta = 8/3$, initial condition $(1, 1, 1)$, and $dt = 0.01$.

2.2 Rössler

A 3D system with a simpler attractor structure:

$$\dot{x} = -y - z \tag{4}$$
$$\dot{y} = x + ay \tag{5}$$
$$\dot{z} = b + z(x - c) \tag{6}$$

with $a = 0.2$, $b = 0.2$, $c = 5.7$, initial $(0.1, 0, 0)$, and $dt = 0.05$.

2.3 Mackey-Glass

A scalar delay-differential equation exhibiting chaotic dynamics for large delay:

$$\dot{x}(t) = \beta \frac{x(t-\tau)}{1 + x(t-\tau)^n} - \gamma x(t) \tag{7}$$

with $\beta = 0.2$, $\gamma = 0.1$, $n = 10$, $\tau = 17$, initial history $x = 0.5$, and $dt = 0.1$. Integrated via Euler with a discrete delay buffer of length τ.

2.4 Lorenz-96

A variable-dimension system used for dimensionality scaling:

$$\dot{x}_i = (x_{i+1} - x_{i-2})x_{i-1} - x_i + F, \quad i = 0, \ldots, N-1 \tag{8}$$

with periodic boundary conditions, $F = 8.0$, $dt = 0.01$, and initial condition $x_i = 8.0$ for all i except $x_{N-1} = 8.01$. The dimension N is varied from 5 (baseline) through 8, 12, 16, 20, and 24.

2.5 Logistic Map

A discrete 1D chaotic map:

$$x_{n+1} = rx_n(1 - x_n) \tag{9}$$

with $r = 4.0$ (fully chaotic), $x_0 = 0.123$.

3 Experimental Configuration

All experiments use identical CLA parameters:

Parameter	Value
Steps	256
Discard (transient)	64
Bins per axis	4
Max iterations	8
N-gram range	2–5
Min uses	2
Categories	Enabled

All trajectories are deterministic; no stochastic RNG is used in the integrators. The only randomness is in the CLA grammar-learning seed, which controls tie-breaking in the MDL greedy loop.

4 Results

4.1 Baseline Benchmarks

System	Dim	Unique	Rules	Score (bits)	Bits/sym
Mackey-Glass	1	4	6	44.4	0.17
Logistic Map	1	4	8	113.2	0.44
Lorenz-63	3	21	8	187.2	0.73
Rössler	3	24	8	189.2	0.74
Lorenz-96	5	56	8	234.2	0.92

Key observations:

- Exact reconstruction holds for all five systems.
- Mackey-Glass achieves the best compression (0.17 bits/symbol) due to its highly repetitive scalar trajectory with only 4 unique symbols.
- The logistic map, also 1D, shows moderate compressibility (0.44 bits/symbol) with more diverse symbol sequences.
- Lorenz-63 and Rössler are closely matched ($\sim$0.74 bits/symbol), consistent with their similar attractor complexity.
- Lorenz-96 at dim=5 already shows high symbol diversity (56 unique out of 256), yielding near-logarithmic compression.

4.2 Dimensionality Scaling (Lorenz-96)

Dim	Unique	Possible (4^N)	Rules	Score (bits)	Bits/sym
5	56	1,024	8	234.2	0.92
8	78	65,536	8	229.2	0.90
12	107	1.7×10^7	6	241.4	0.94
16	121	4.3×10^9	3	248.7	0.97
20	127	1.1×10^{12}	6	251.4	0.98
24	148	2.8×10^{14}	5	251.5	0.98

Key observations:

- Exact reconstruction holds at all tested dimensions (1–24).
- Unique symbol count grows sub-exponentially relative to the theoretical maximum 4^N, indicating the trajectory occupies a small fraction of partition space.
- Compression degrades monotonically: from 0.92 bits/symbol at dim=5 to 0.98 at dim=24, approaching the entropy of a flat uniform distribution over unique symbols.
- At dim=16 (matching OmegaSim's 16-dimensional state vector), CLA discovers only 3 chunk rules and 121 unique symbols, indicating a near-incompressible regime with 4 bins and 256 steps.
- At dim=20–24, the system is effectively incompressible at this bin/step resolution. This is the regime where dimension reduction or coarser binning becomes essential.

5 OmegaSim Dimensionality Context

The OmegaSim A6 thresholded-appraisal model has the following continuous state vector:

Component	Dimensions	Range
Motivation (4 roles)	4	$[0, 1]$
Fatigue (4 roles)	4	$[0, 1]$
Threshold (4 roles)	4	$[0.15, 0.85]$
Artifact maturity	1	$[0, 1]$
Provenance debt	1	$[0, 1]$
Risk	1	$[0, 1]$
Prediction error	1	$[0, 1]$
Total continuous	**16**	

An additional discrete field (`last_action`, $\in \{0, 1, 2, 3\}$) is not symbolized as a continuous variable.

The dimensionality scaling results show that at $N = 16$, the M1 symbolization with 4 bins per axis already enters the near-incompressible regime. This has two implications for the CLA→OmegaSim bridge:

1. **Direct M1 symbolization at $N = 16$ is feasible but marginal.** CLA maintains exact reconstruction and still discovers a small number of chunk rules, but the grammar captures only the most repetitive subsequences. The symbolic vocabulary (121 unique symbols out of $4^{16} \approx 4.3 \times 10^9$) is sparse but diverse enough that MDL gains are limited.

2. **Dimension reduction or selective symbolization is needed for richer grammar discovery.** Candidate approaches include:

    - Symbolizing only a subset of state variables (e.g., the 4 motivation values, which are the primary behavioral output).
    - Applying PCA or delay-embedding before symbolization.
    - Using coarser bins (2–3 per axis) to reduce symbol diversity.
    - Employing a sliding-window or delay-coordinate symbolization rather than per-timestep partitioning.

6 Implementation Details

6.1 Repository

- GitHub: `bgoertzel-sing/chaos-language-algorithm`
- Package: `chaoslang` (pure Python, no external dependencies)
- Latest commit: `847390c` (sprint complete)
- Tests: 64 passing

6.2 Sprint Completion Status

All five sprint tasks are complete:

Task	Description	Status
1	Symbolic-string MVP (chunks, categories, MDL)	✓
2	Recorded benchmark experiments (5 systems)	✓
3	Dimensionality scaling (Lorenz-96 dims 8–24)	✓
4	JS-divergence context clustering	✓
5	Persistence (JSON state, edit-log replay, stable text)	✓

6.3 CLI Usage

```
# Single system benchmark
PYTHONPATH=src python3 -m chaoslang.benchmarks.m1_controls \
    --system lorenz63 --steps 256 --discard 64 --bins 4 --iterations 8

# Lorenz-96 with custom dimension
PYTHONPATH=src python3 -m chaoslang.benchmarks.m1_controls \
    --system lorenz96 --steps 256 --discard 64 --bins 4 \
    --iterations 8 --dimension 16 --seed 0
```

7 Conclusions and Next Steps

1. CLA successfully discovers grammar structure in all five standard attractor systems with exact reconstruction, validating the core algorithm.

2. The dimensionality scaling experiments reveal a clear transition from compressible (low-dim) to near-incompressible (dim $\geq$ 16) regimes at the current M1 resolution of 4 bins and 256 steps.

3. The OmegaSim A6 model's 16-dimensional state vector sits exactly at this transition boundary, making it the natural test case for bridging textbook benchmarks to simulation-scale traces.

4. The next step is to select toy attractor systems at $N \approx 16$ and test dimension-reduction strategies (selective symbolization, PCA, delay-embedding, coarser binning) to improve grammar discovery in this regime.

5. Once CLA can robustly extract grammar from $N \approx 16$ systems, it can be applied to actual OmegaSim traces to assess whether the thresholded-appraisal dynamics produce complex strange-attractor structure.